Clustering

↳ an unsupervised learning technique used to divide large sets of unlabelled data into logical groups.

grouping is based on Characteristics of the data.

→ clustering algorithms examine all data attributes to discover hidden patterns and natural groupings.

→ K-Means clustering

↳ used on data that does not have predefined outputs or labels

→ Find similarities between unorganized data points and group them together into clusters.

Elbow method

↳ visual technique used to figure out
exactly what k value should be

→ decide the optimal number of
clusters to create.

WCSS → within Cluster Sum of squares.

→ measures the distance between each
data point and cluster centroid.
Then square the distance

number of clusters ↑↑
↳ WCSS ↓↓ .

Inliers → data points located near the center of a cluster and far from other clusters

PCA → Principal component Analysis.

Simplifies massive data sets by reducing the number of features

identifies the directions in the data that capture the most variation.

Evaluation metrics

↳ required to mathematically determine how well a clustering model performs

→ Inertia

→ Dunn Index